

Lecture 5 – Abstract Interpretation

SWS128: Software Analysis

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2026 Fall

- A **dataflow analysis**: data and a join, a transfer per edge, a direction, an initial value. One engine, `solve`, runs them all.
- We said its answer is **sound** and the **least** one. We never said why.

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- We said its answer is **sound** and the **least** one. We never said why.
- Today: the theory behind those two words. Dataflow analysis turns out to be one instance of it.

1. From Concrete to Abstract

Collecting Semantics

Abstract Domains

2. Two Analyses

Step by Step

3. Galois Connections

The Connection

Transformers

4. Fixpoints

Existence

Computation

5. In Python

Code

Running It

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Running It

```
def f(c):  
    x = 1  
    if c < 0:  
        x = -1  
    y = x + 2  
    return y
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One run per input:

c	returns
-3	1
0	3
5	3
$\vdots$	$\vdots$

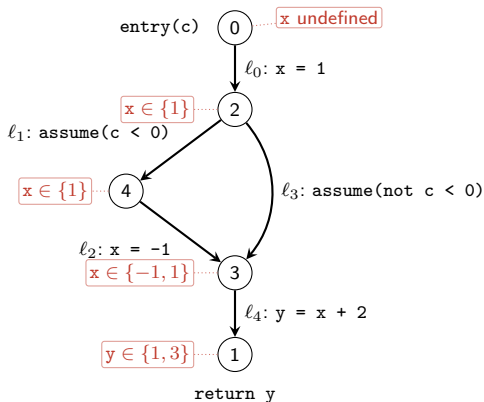
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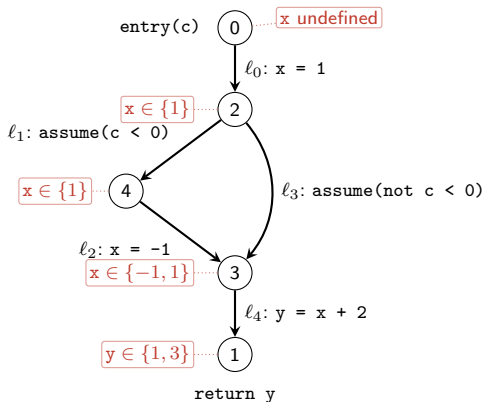
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The interpreter answers for **one** c . An analysis must answer for **every** c at once: *is y always positive?*

At each point, the values x and y can have, over **all** runs:



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The exact answer: y is always positive. The equations are the dataflow ones, $C(q) = \bigcup_{p \rightarrow q} f(C(p))$, over sets of concrete values.

With a Loop, It Never Stops

```
def g(n):  
    x = 0  
    while x < n:  
        x = x + 1  
    return x
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The values of x at the loop head, one iteration of the equations at a time:

$$\{0\} \subset \{0, 1\} \subset \{0, 1, 2\} \subset \dots$$

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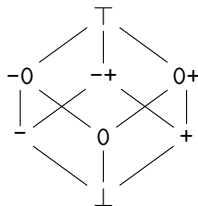
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- The chain never stabilizes: n can be any number, so the exact set is **infinite**, and no finite number of iterations reaches it.
- The collecting semantics is exact but, in general, **cannot be computed**. We need a **finite** description of each set.

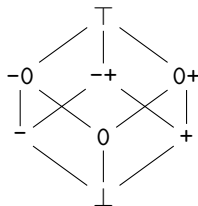
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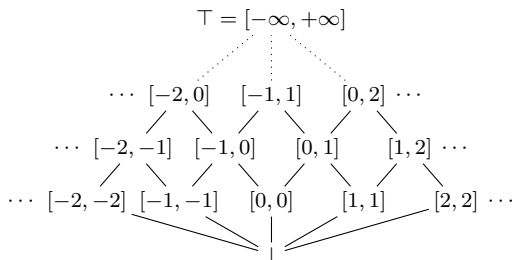


Sign: which of $-$, 0 , $+$ can occur

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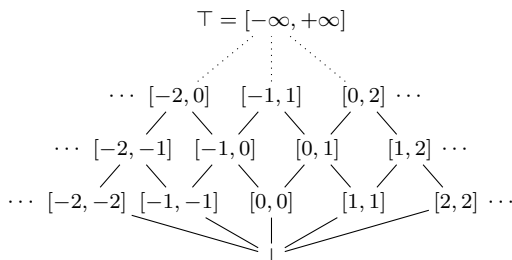
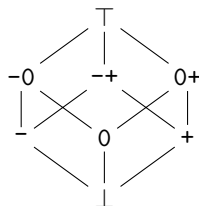


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Interval: $[l, u]$, ordered by inclusion

Replace each set of values by a **finite description** of it. We draw each domain as a **Hasse diagram**: a line goes up from smaller to larger.



Sign: which of $-$, 0 , $+$ can occur

Interval: $[l, u]$, ordered by inclusion

Set of values	As a sign	As an interval
$\{1\}$	$+$	$[1, 1]$
$\{-1, 1\}$	$-+$	$[-1, 1]$
$\{0, 1, 2, \dots\}$	$0+$	$[0, +\infty]$

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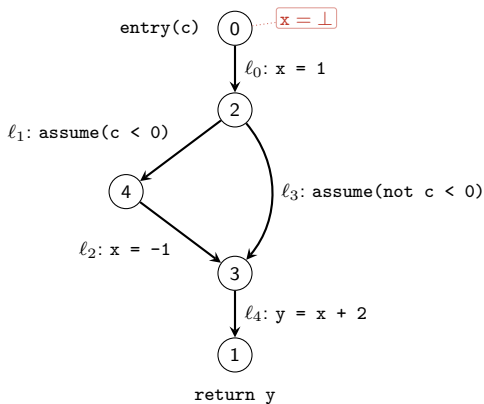
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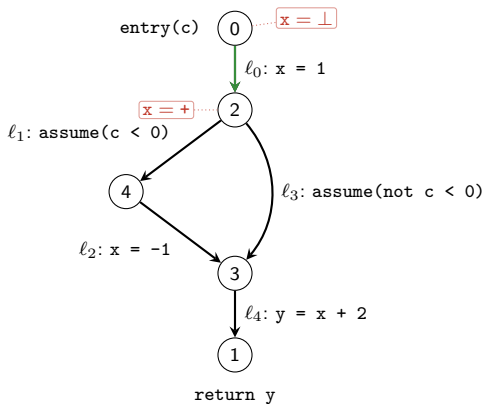
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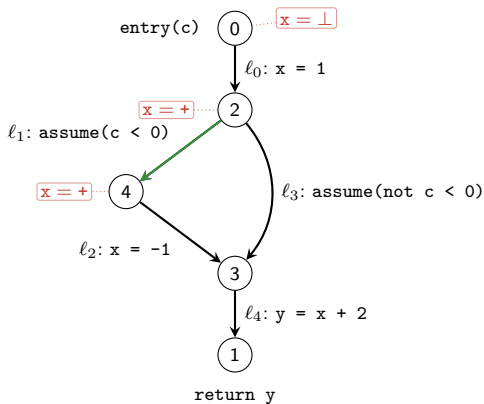
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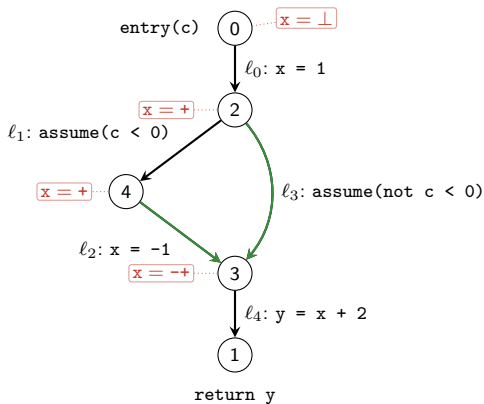
Initial: x has no value yet, $\perp$.



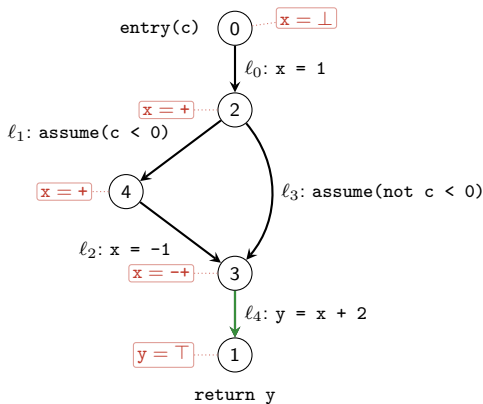
$\ell_0: x = 1$, and 1 is positive: +.



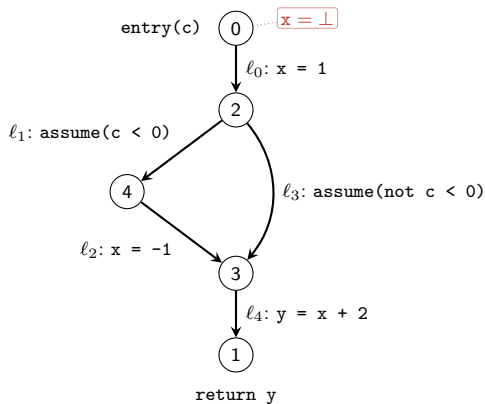
$l_1: \text{assume}(c < 0)$ says nothing about x .



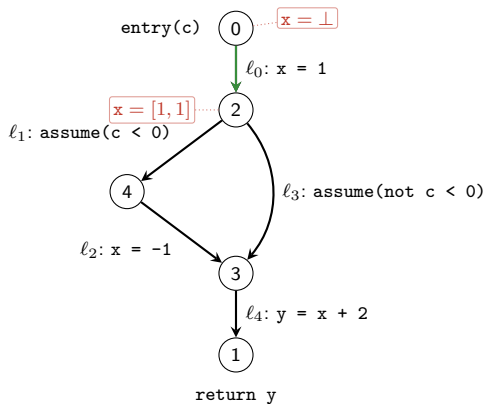
l_2 gives $-$, l_3 gives $+$. **Join:** $-\perp+ = -+$.



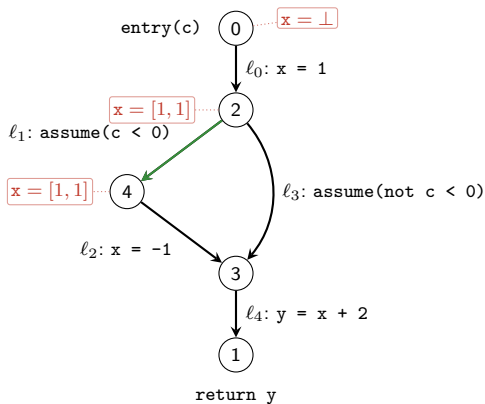
ℓ_4 : adding + to $-+$ gives $y = \top$, since $-1 + 2 > 0$ but $-5 + 2 < 0$. Is y positive? Sign **cannot tell**.



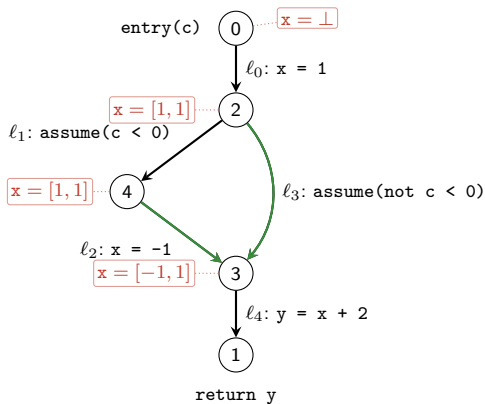
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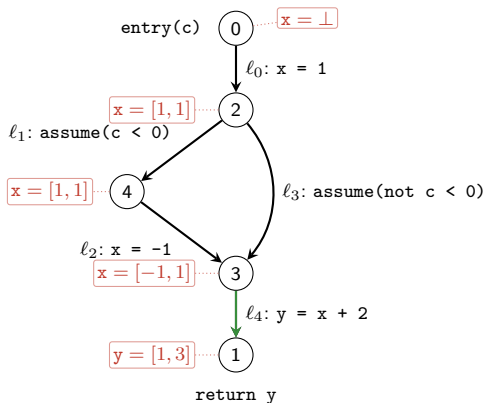
$\ell_0: x = 1: [1, 1].$



ℓ₁: x unchanged. (c becomes $[-\infty, -1]$ here.)



ℓ_2 gives $[-1, -1]$, ℓ_3 gives $[1, 1]$. **Join:** $[-1, 1]$.



ℓ_4 : adding $[2, 2]$ to $[-1, 1]$ gives $y = [1, 3]$. *Is y positive?* Interval says **yes**.

	concrete	sign	interval
x at 3	$\{-1, 1\}$	$-+$	$[-1, 1]$
y at 1	$\{1, 3\}$	$\top$	$[1, 3]$
y positive?	yes	<i>don't know</i>	yes

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y at 1	$\{1, 3\}$	$\top$	$[1, 3]$
y positive?	yes	<i>don't know</i>	yes

- Both are **correct**: every real value is inside what they say.
- Sign kept “not zero” but not *how big*: a negative plus 2 can be anything. Interval kept $[-1, 1]$, enough to add 2 and stay positive.
- Choosing a domain is choosing what you can prove.

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Two functions connect sets of values and the abstract domain:

- $\gamma(a)$: the set of values a describes.

$$\gamma(+) = \{1, 2, 3, \dots\} \quad \gamma([1, 3]) = \{1, 2, 3\}$$

- $\alpha(S)$: the best description of S .

$$\alpha(\{-1, 1\}) = -+ \text{ (sign)} \quad \alpha(\{-1, 1\}) = [-1, 1] \text{ (interval)}$$

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They form a **Galois connection** when

$$\alpha(S) \sqsubseteq a \iff S \subseteq \gamma(a)$$

that is, a describes S correctly exactly when a is above $\alpha(S)$.

$$S = \{1, 100\} \quad \alpha(S) = [1, 100] \quad \gamma(\alpha(S)) = \{1, 2, \dots, 100\}$$

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- Intervals cannot say “1 or 100”. They say “between 1 and 100”, and 98 values that never occur come with it.
- Always $S \subseteq \gamma(\alpha(S))$: abstraction adds values, never removes them. That is the price of computability.

Each operation needs an abstract version. For sign, define it on the three signs:

$+\#$	$-$	0	$+$
$-$	$-$	$-$	$\top$
0	$-$	0	$+$
$+$	$\top$	$+$	$+$

$\times\#$	$-$	0	$+$
$-$	$+$	0	$-$
0	0	0	0
$+$	$-$	0	$+$

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$+^\#$	-	0	+
-	-	-	$\top$
0	-	0	+
+	$\top$	+	+

$\times^\#$	-	0	+
-	+	0	-
0	0	0	0
+	-	0	+

- Any other element is a set of signs: take every pair, then join.

$$-+ +^\# + = (-+^\# +) \sqcup (+ +^\# +) = \top \sqcup + = \top$$

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0	-	0	+
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-	+	0	-
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- Sound:** $f(\gamma(a)) \subseteq \gamma(f^\#(a))$. A negative times a negative is always positive, so $- \times^\# - = +$.
- Interval: $[a, b] +^\# [c, d] = [a + c, b + d]$.

The analysis is **sound** when, at every point,

$$C(n) \subseteq \gamma(A(n))$$

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	C	$\gamma(\text{sign})$	$\gamma(\text{interval})$
x at 2	$\{1\}$	$\{1, 2, \dots\}$ ✓	$\{1\}$ ✓
x at 3	$\{-1, 1\}$	$\mathbb{Z} \setminus \{0\}$ ✓	$\{-1, 0, 1\}$ ✓
y at 1	$\{1, 3\}$	$\mathbb{Z}$ ✓	$\{1, 2, 3\}$ ✓

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$$C(n) \subseteq \gamma(A(n))$$

	C	$\gamma(\text{sign})$	$\gamma(\text{interval})$
x at 2	$\{1\}$	$\{1, 2, \dots\} \checkmark$	$\{1\} \checkmark$
x at 3	$\{-1, 1\}$	$\mathbb{Z} \setminus \{0\} \checkmark$	$\{-1, 0, 1\} \checkmark$
y at 1	$\{1, 3\}$	$\mathbb{Z} \checkmark$	$\{1, 2, 3\} \checkmark$

- Each transfer is sound and each join is sound, so the result is.
- This is the contract of every analysis in this course.

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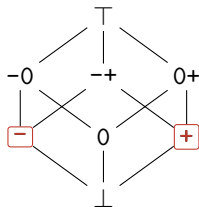
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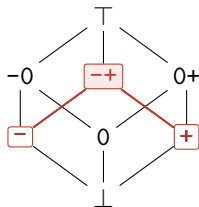
Running It

Pick any elements. Is there a **lowest** element above all of them?



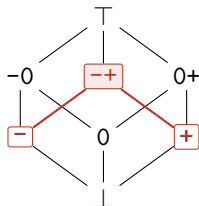
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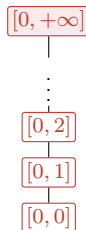


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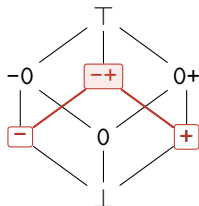


Lowest above $-$ and $+$: $-+$

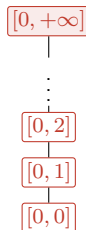


Lowest above $[0, n]$ for all n : $[0, +\infty]$

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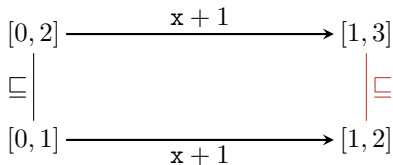
- A **complete lattice**: the answer is always yes. The lowest element above a set is its **join** $\sqcup$.

F is **monotone** if a larger input never gives a smaller output:

$$a \sqsubseteq b \implies F(a) \sqsubseteq F(b)$$

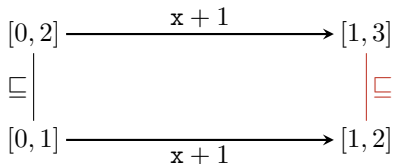
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- Every transformer we built is monotone: $+\sharp$, $\times\sharp$, and $\sqcup$. So is F , which only combines them.

All the equations together are one function F on the table of abstract values. The answer is a table A with $F(A) = A$.

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Theorem (Knaster–Tarski)

If $\mathbb{D}$ is a complete lattice and F is monotone, then F has a **least fixpoint**, $\text{lfp } F$.

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Theorem (Knaster–Tarski)

If $\mathbb{D}$ is a complete lattice and F is monotone, then F has a **least fixpoint**, $\text{lfp } F$.

- Every fixpoint is sound. The least one is the most precise.
- So the answer we want exists, and there is exactly one.

Start from $\perp$ and apply F until nothing changes:

$$\perp \sqsubseteq F(\perp) \sqsubseteq F^2(\perp) \sqsubseteq \dots \quad \mathbf{lfp} F = \bigsqcup_i F^i(\perp)$$

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x at the loop head of g , in sign:

	F^0	F^1	F^2	F^3
x	$\perp$	0	$0 \sqcup (0 +^\# +) = 0+$	$0 \sqcup (0+ +^\# +) = 0+$

Start from $\perp$ and apply F until nothing changes:

$$\perp \sqsubseteq F(\perp) \sqsubseteq F^2(\perp) \sqsubseteq \dots \quad \text{fix } F = \bigsqcup_i F^i(\perp)$$

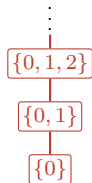
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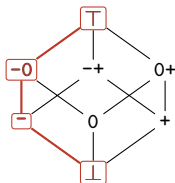
- Two steps and it is stable at $0+$. The concrete chain $\{0\}, \{0, 1\}, \dots$ never was.
- The worklist computes this same chain: same fixpoint, less work.

The **height** of a domain is the length of its longest chain $a_0 \sqsubset a_1 \sqsubset \dots$.

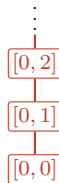
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Concrete: infinite



Sign: 3

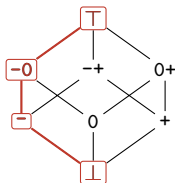


Interval: infinite

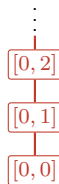
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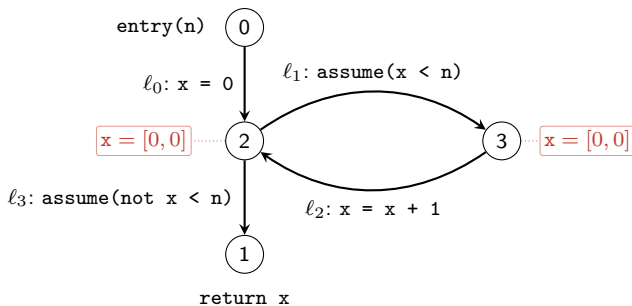
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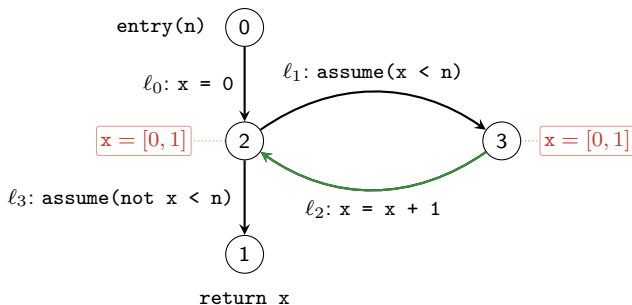
- Each Kleene step either stays or moves up. With finite height, it must stop. Sign is guaranteed to terminate.
- Concrete and Interval have an infinite chain: **no guarantee**.

The same loop g, in interval:



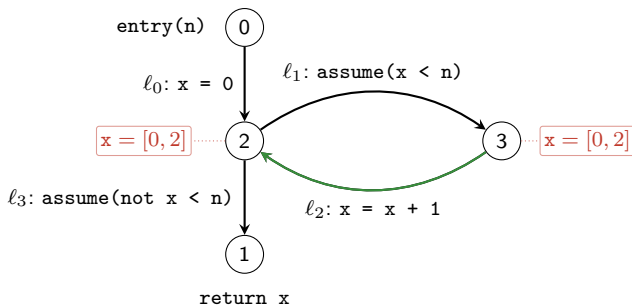
$\ell_0: x$ starts at $[0, 0]$.

The same loop g, in interval:



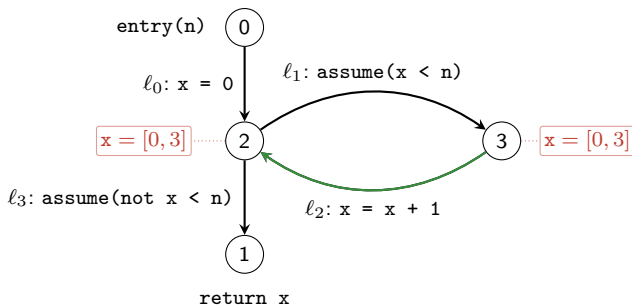
Around the loop: $[0, 0] +^\# [1, 1] = [1, 1]$. **Join** at 2: $[0, 1]$.

The same loop g, in interval:



Around again: $[0, 1] +^\# [1, 1] = [1, 2]$. Join: $[0, 2]$.

The same loop g , in interval:



Each lap adds one: $[0, 0] \subseteq [0, 1] \subseteq [0, 2] \subseteq \dots$ **never stops**.

- Sign stopped at $0+$ after two steps. Interval has **infinite height**: the chain has no end.
- The engine gives up (`RuntimeError: no fixpoint`). The fix, **widening**, is the next lecture.

Abstract interpretation	Dataflow analysis
abstract domain $\mathbb{D}$	sets of facts, ordered by $\subseteq$
abstract transformer $f^\sharp$	$\text{gen} \cup (X \setminus \text{kill})$
join $\sqcup$	$\cup$ or $\cap$
$\text{lfp } F$	what <code>solve</code> returned
soundness $C \subseteq \gamma(A)$	assumed, never stated

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- A recipe for a new analysis: pick $\mathbb{D}$, define γ , write a sound $f^\sharp$ per operation, run the engine.

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```
class Domain:
    bot: object
    top: object

    def le(self, a, b) -> bool: raise NotImplementedError
    def join(self, a, b): raise NotImplementedError
    def meet(self, a, b): raise NotImplementedError
    def alpha(self, ns: set[int]): raise NotImplementedError
    def gamma(self, a) -> str: raise NotImplementedError
    def of_num(self, n: int): raise NotImplementedError
    def binop(self, op: str, a, b): raise NotImplementedError
    def unop(self, op: str, a): raise NotImplementedError

    def widen(self, a, b): return self.join(a, b)
    def narrow(self, a, b): return self.meet(a, b)
```

- The lattice (`le`, `join`, `meet`), the Galois connection (`alpha`, `gamma`), and the transformers (`of_num`, `binop`, `unop`).
- `widen` and `narrow` are for the next lecture.

```
# an element is a set of the three signs; its name is the key
ATOMS = {BOT: "", NEG: "-", ZERO: "0", POS: "+",
         NONPOS: "-0", NONNEG: "0+", NONZERO: "-+", TOP: "-0+"}
NAME = {frozenset(v): k for k, v in ATOMS.items()}

ADD = {("-", "-"): "-", ("-", "0"): "-", ("-", "+"): "-0+",
       ("0", "0"): "0", ("0", "+"): "+", ("+", "+"): "+"}

class Sign(Domain):
    def join(self, a, b):
        return NAME[frozenset(ATOMS[a] + ATOMS[b])]
    def binop(self, op, a, b):
        ...
        out = ""
        for x in ATOMS[a]:
            for y in ATOMS[b]:
                out += table.get((x, y)) or table[(y, x)]
        return NAME[frozenset(out)]
```

- Join is union of signs. An operation is the three-sign table, applied to every pair.

```
>>> from syntax import parse
>>> from cfg import build
>>> from domain import Sign, analyze, returned
>>> src = """
... def f(c):
...     x = 1
...     if c < 0:
...         x = -1
...     y = x + 2
...     return y
... """
>>> g = build(parse(src))
>>> d = Sign()
>>> d.join("-", "+"), d.binop("+", "-+", "+")
('+-+', 'top')
>>> res = analyze(g, d)
>>> res[3]["x"], returned(g, d, res)
('+-+', 'top')
```

- The same answers as the step-by-step: x is $-+$ at the merge, y is $\top$.
The interval domain is the next homework.

1. From Concrete to Abstract

Collecting Semantics

Abstract Domains

2. Two Analyses

Step by Step

3. Galois Connections

The Connection

Transformers

4. Fixpoints

Existence

Computation

5. In Python

Code

Running It

- Widening and Narrowing

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